

Swastika Khatua

Computer Science Engineering Student

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Professional Summary

Computer Science Engineering student with strong fundamentals in machine learning and generative AI. Focused on agentic AI, with understanding of Retrieval-Augmented Generation (RAG), multi-step reasoning, and LLM orchestration, and hands-on experience building end-to-end ML and GenAI pipelines using Python, LangChain, vector databases, and FastAPI.

Education

- **Bachelor of Technology (BTech) in Computer Science and Engineering** Aug 2023 – Present
KIIT University CGPA: 9.15 / 10
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming
- **Holy Cross School**
ISC (Class XII): 94.75% 2021–2023
ICSE (Class X): 98.6% 2020

Skills

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|--------------------------------|-------------------------------------|----------------|
| • Python | • Deep Learning | • LangGraph |
| • Java | • Natural Language Processing (NLP) | • Streamlit |
| • C (Basics) | • Ollama | • Docker |
| • Data Structures & Algorithms | • RAG | • FastAPI |
| • Object-Oriented Programming | • Scikit-learn | • Git & GitHub |
| • MySQL | • TensorFlow (Keras) | • LangChain |
| • Machine Learning | • PyTorch | • GenAI |

Projects

- **Hybrid Movie Recommender System (ML – NLP)** 📄
Built a hybrid recommendation system combining **content-based NLP techniques** (TF-IDF, cosine similarity) with **collaborative filtering** to generate personalized movie recommendations. Deployed model inference using **FastAPI REST APIs**.
- **Retrieval Augmented Generation (RAG) Research Paper Summarizer (GenAI – NLP)** 📄
Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain**, **ChromaDB**, and **Google Gemini** to generate context-grounded summaries from documents. Leveraged **semantic embeddings** and vector similarity search to reduce hallucinations and improve factual accuracy.
- **Next Word Prediction (Deep Learning – NLP)** 📄
Built a **BiLSTM-based language model** with attention mechanism and POS embeddings for next-word prediction. Deployed the trained model using **FastAPI** with a **Streamlit frontend** for interactive inference.

Certifications

- **Introduction to Deep Learning & Neural Networks with Keras** (IBM – Coursera, Jan 2026) 📄
- **Oracle Cloud Infrastructure 2023 Certified Foundations Associate** (Oct 2025 – Oct 2027) 📄
- **Oracle Cloud Infrastructure 2023 Certified AI Foundations Associate** (Oct 2025 – Oct 2027) 📄
- **Oracle Cloud Infrastructure 2023 Certified Generative AI Professional** (Oct 2025 – Oct 2027) 📄

Experience

- **Digital Marketing Intern, Veedol** (15 May 2025 – 28 June 2025) — Conducted **data-driven analysis** of marketing campaigns, improving engagement by 15%.

Personal Skills

- Strong analytical mindset with a structured approach to problem-solving
- Effective team collaborator in academic and professional environments
- Clear and concise verbal and written communication skills
- Adaptable quick learner with strong time management and multitasking skills
- Languages: English (Professional), Hindi (Professional), Bengali (Native)